Oracle Utilities Service and Measurement Data Foundation v25.10HOT

Product Fix Design

Bug 39109281

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Oracle Utilities Service and Measurement Data Foundation Product Fix Design

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Document History

Date	Name	Description/Reason for change
2026-03-23	Mahesh Burgu	Bug 39109281 - One Day Closing Bill Fails on Eligibility Criteria using algorithm type D1-RCVEURCF

Review History

Date	Reviewer Name	Notes
2026-03-23	Mahesh Burgu	Final Review Completed

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Reported Issues

Description

One Day Closing Bill Fails on Eligibility Criteria using algorithm type D1-RCVEURCF throwing an error 'No Service points are linked to usage subscription'.

Steps to Recreate

1. Create a device, device configuration, interval MC.
2. Create a service point and install the device on service point.
3. Create measurements on the interval MC from 03-30-2018 12 A.M to 07-25-2018 12:00AM.
4. Create usage calculation group with D2-Math Usage rule.
    a. **Add “Usage Calculation Rule Eligibility Criteria” using the Algorithm type D1-RCVEURCF (Retrieve Characteristic Value for Evaluation - Usage Rule EC Field).**
5. Create usage subscription with above service point and usage calculation group.
6. Move the usage subscription to inactive status and set the end date/time as 07-24-2018 12 A.M.
7. Create a usage transaction for the following period -

    Start date/time 07-24-2018 12 A.M and End date/time 07-25-2018 12 A.M.
8. Observe that usage transaction is transitioned to Issue Detected state with the error “No Service Points are linked to Usage Subscription.”

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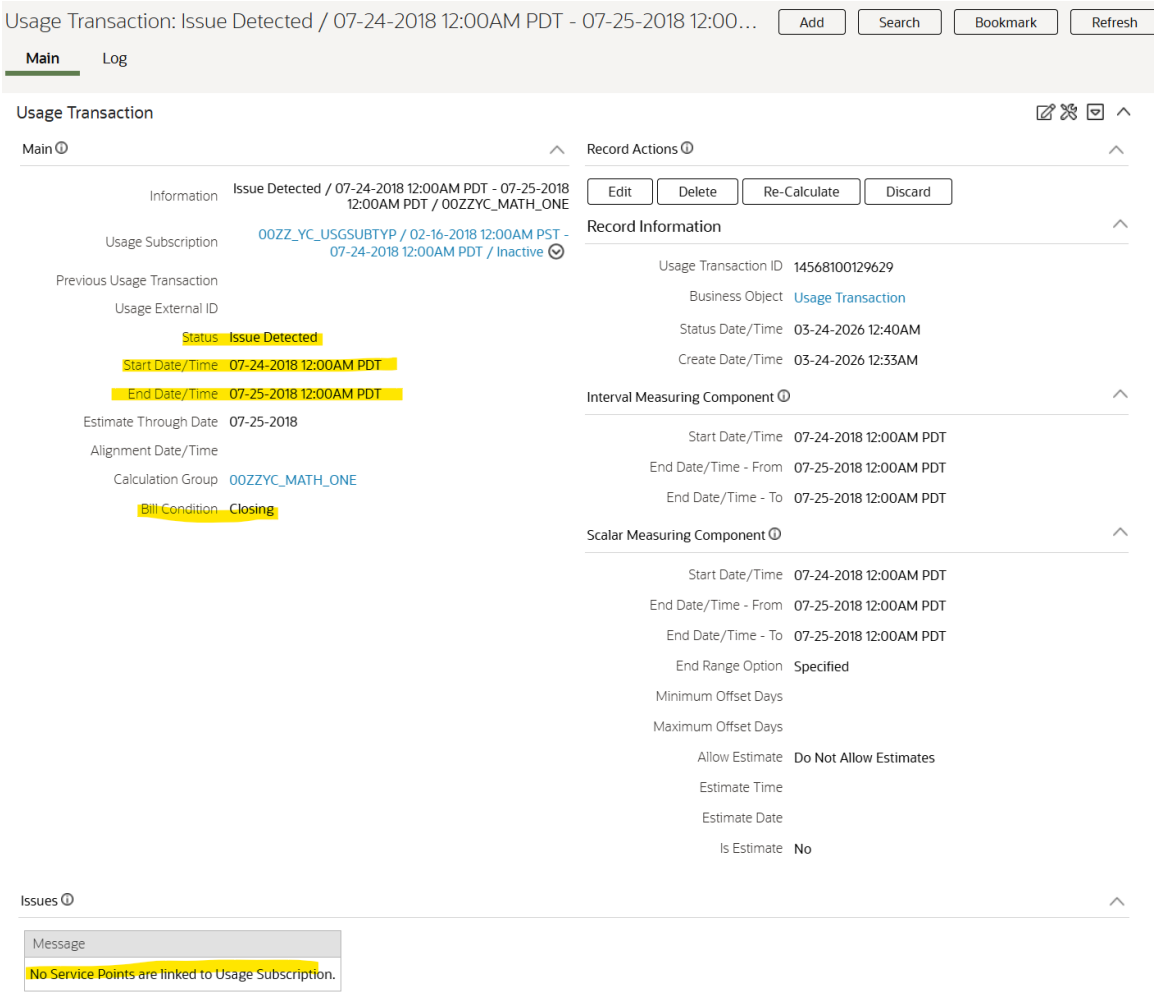

Investigation

Problem Cause

During the usage calculation for the usage rule configured using the Algorithm type D1-RCVEURCF (Retrieve Characteristic Value for Evaluation - Usage Rule EC Field), service points linked to the usage subscription are retrieved. Only the service points whose stop date/time is greater than usage transaction start date/time are considered. Due to this usage for single day closing bill, where the usage transaction starts on the usage subscription end date/time, is not calculated.

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Suggested Solution

In UsageSubscription_Impl.java, make changes to the method retrieveServicePointsAsOfDateTime(DateTime referenceDateTime) such that the service points linked to the usage subscription are considered for further usage calculation when usage transaction starts on the usage subscription end date/time (assuming it as a Closing bill condition).

Solution Design

In UsageSubscription_Impl.java, make changes to the method retrieveServicePointsAsOfDateTime(DateTime referenceDateTime) such that the service points linked to the usage subscription are considered for further usage calculation when usage transaction starts on the usage subscription end date/time (assuming it as a Closing bill

Test Case 1: Verify that usage is calculated for a single day closing usage transaction when the usage transaction starts on the usage subscription stop date/time and D2-Math Usage rule is used.

1. Create a device, device configuration, interval MC.
2. Create a service point and install the device on service point.
3. Create measurements on the interval MC from 03-30-2018 12 A.M to 07-25-2018 12:00AM.
4. Create usage calculation group with D2-Math Usage rule.
    a. **Add “Usage Calculation Rule Eligibility Criteria” using the Algorithm type D1-RCVEURCF (Retrieve Characteristic Value for Evaluation - Usage Rule EC Field).**
5. Create usage subscription with above service point and usage calculation group.
6. Move the usage subscription to inactive status and set the end date/time as 07-24-2018 12 A.M.
7. Create a usage transaction for the following period -

    Start date/time 07-24-2018 12 A.M and End date/time 07-25-2018 12 A.M.

    Bill condition - closing
8. Verify that usage is calculated on the usage transaction.

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Test Case 2: Verify that usage is calculated for a closing usage transaction that is created for two days.

1. Create a device, device configuration, interval MC.
2. Create a service point and install the device on service point.
3. Create measurements on the interval MC from 03-30-2018 12 A.M to 07-25-2018 12:00AM.
4. Create usage calculation group with D2-Math Usage rule.
    a. **Add "Usage Calculation Rule Eligibility Criteria" using the Algorithm type D1-RCVEURCF (Retrieve Characteristic Value for Evaluation - Usage Rule EC Field).**
5. Create usage subscription with above service point and usage calculation group.
6. Move the usage subscription to inactive status and set the end date/time as 07-24-2018 12 A.M.
7. Create a usage transaction for the following period -

    Start date/time 07-23-2018 12 A.M and End date/time 07-25-2018 12 A.M.

    Bill condition - closing
8. Verify that usage is calculated on the usage transaction.

Resolution Verification

Test Case 1: Verify that usage is calculated for a single day closing usage transaction when the usage transaction starts on the usage subscription stop date/time and D2-Math Usage rule is used.

1. Create a device, device configuration, interval MC.
2. Create a service point and install the device on service point.

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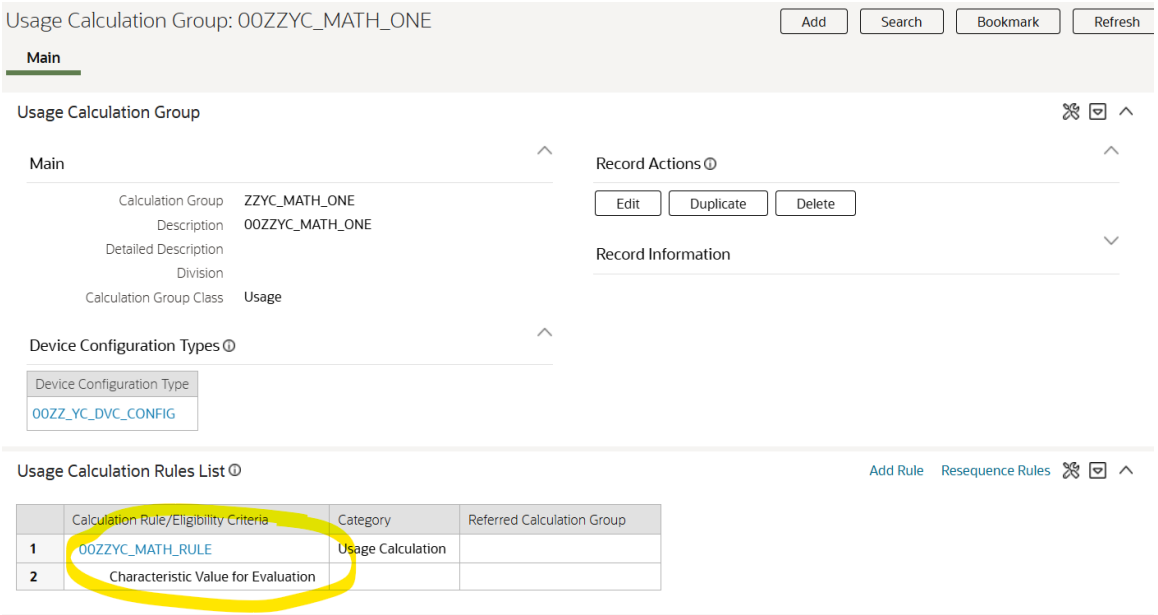
3. Create measurements on the interval MC from 03-30-2018 12 A.M to 07-25-2018 12:00AM.
4. Create usage calculation group with D2-Math Usage rule.



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Usage Calculation Rule

Add | Search | Bookmark | Refresh

Main

Calculation Group	00ZZYC_MATH_ONE
Calculation Rule	ZZYC_MATH_RULE
Sequence	10
Description	00ZZYC_MATH_RULE
Detailed Description	
Calculation Rule Category	Usage Calculation

Interval Vector 1

Type	Physical Channels Linked To Usage Subscription
Unit of Measure	Kilowatt-Hour
Time of Use	
Service Quantity Identifier	
Target Unit Of Measure	
Check Missing Intervals	Yes
SP Device Filter	

Vector Processing

Common Interval Size	01:00:00
Vector Formula Source	Simple Vector Formula
Simple Vector Formula	IV1

Result

Unit of Measure	Kilowatt-Hour
Time of Use	
Service Quantity Identifier	
Insert Primary SQ Entry	Yes
Perform Measurement Quality Assessment	No
SQ Entry Quantity Source	Set Function Against Derived Vector
Set Function Against Derived Vector	Total
Save Derived Vector to Transaction	No
Extract Interval Data	
Apply TOU Map To Derived Vector	No
Condition Code Option	Lowest Quality
Usage Period Type Option	
Enforce US Dynamic Option	No
Condition Code for No Meter	

SQ Rounding Details

Rounding Method	Round Nearest
Decimal Positions	6

Edit | Duplicate | Delete

Record Information

Business Object	Vector and Service Quantity Math

Usage Transaction Exceptions

Usage Calculation Exception

Exception Type	Usage Calculation
Exception Severity	Issues

Measurement Retrieval Exception

Exception Type	Measurement Retrieval
Exception Severity	Issues

Configuration Exception

Exception Type	Configuration
Exception Severity	Issues

Invalid Data Exception

Exception Type	Invalid Data
Exception Severity	Issues

Eligibility Criteria List

Add

	Sequence	Description	Edit	Duplicate	Delete
1	10	Characteristic Value for Evaluation			

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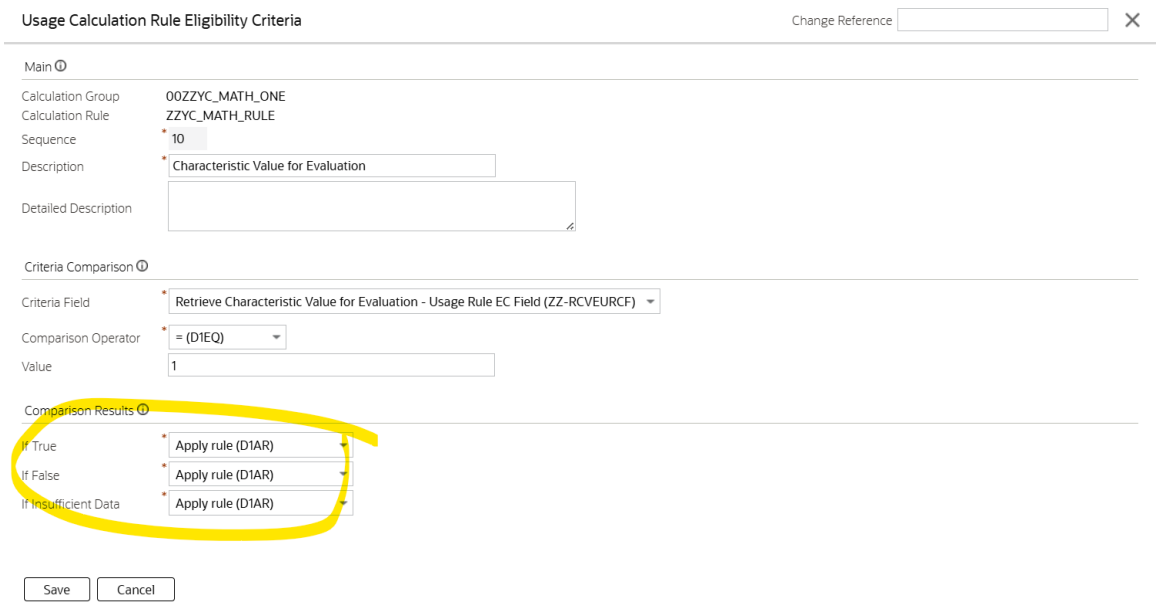
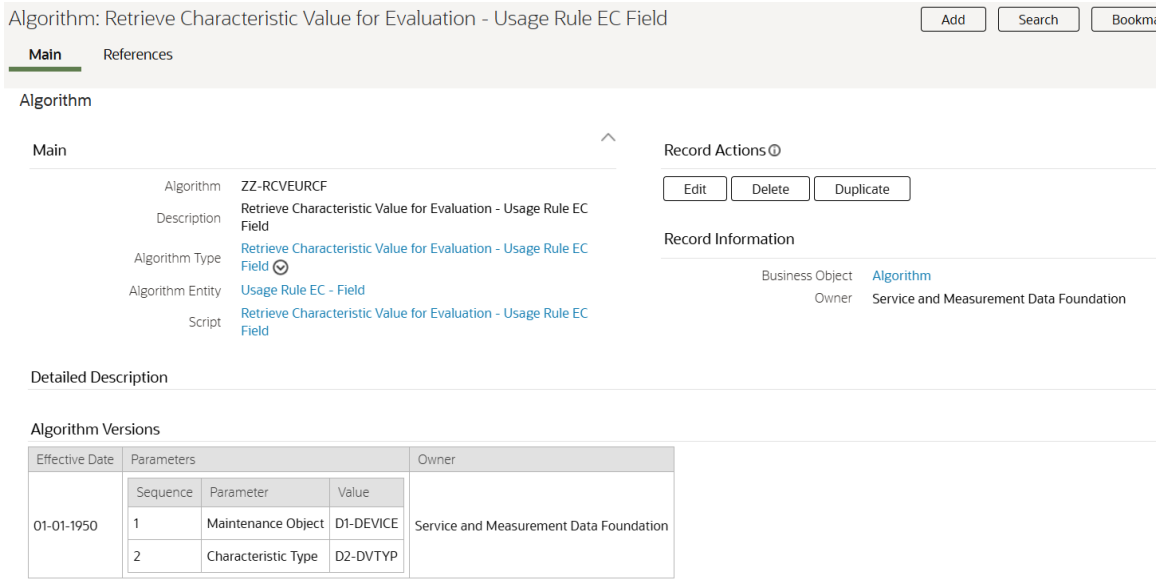


Eligibility Criteria Algorithm



5. Create usage subscription with above service point and usage calculation group.

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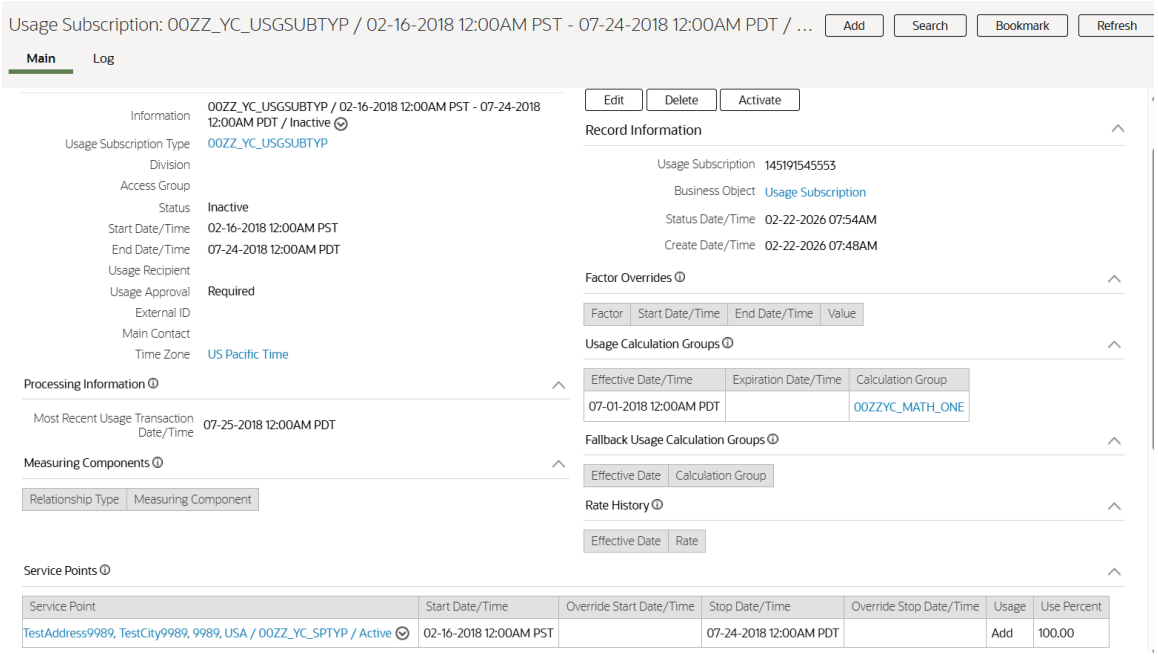
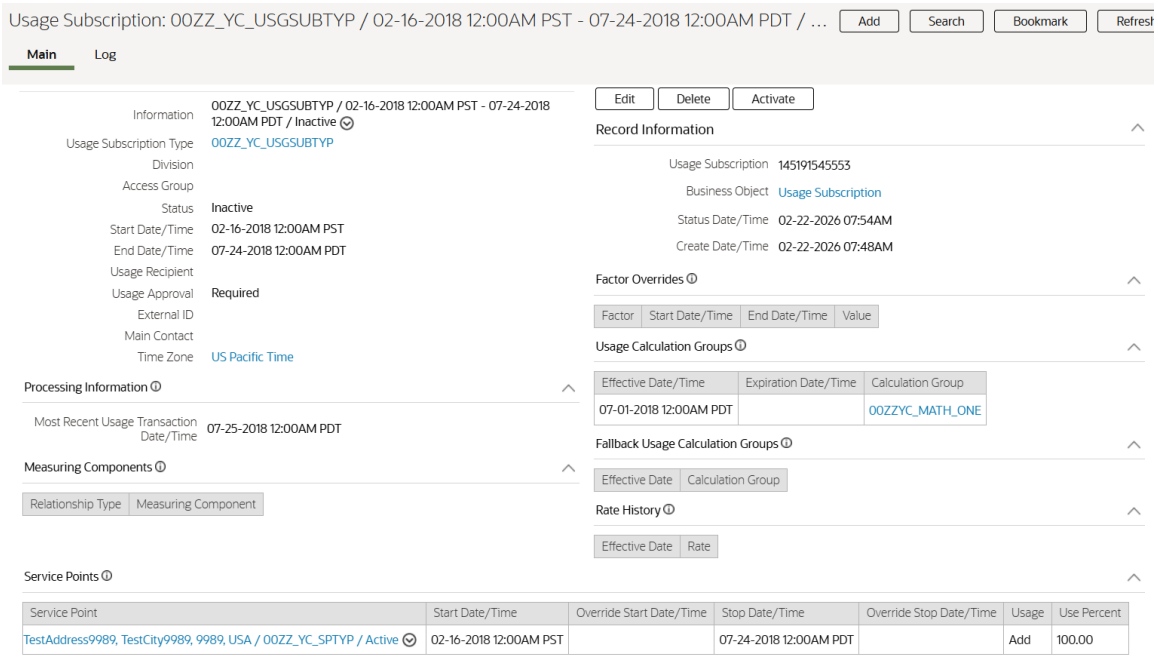


6. Move the usage subscription to inactive status and set the end date/time as 07-24-2018 12 A.M.



7. Create a usage transaction for the following period -

   Start date/time 07-24-2018 12 A.M and End date/time 07-25-2018 12 A.M.

8. Verify that usage is calculated on the usage transaction.

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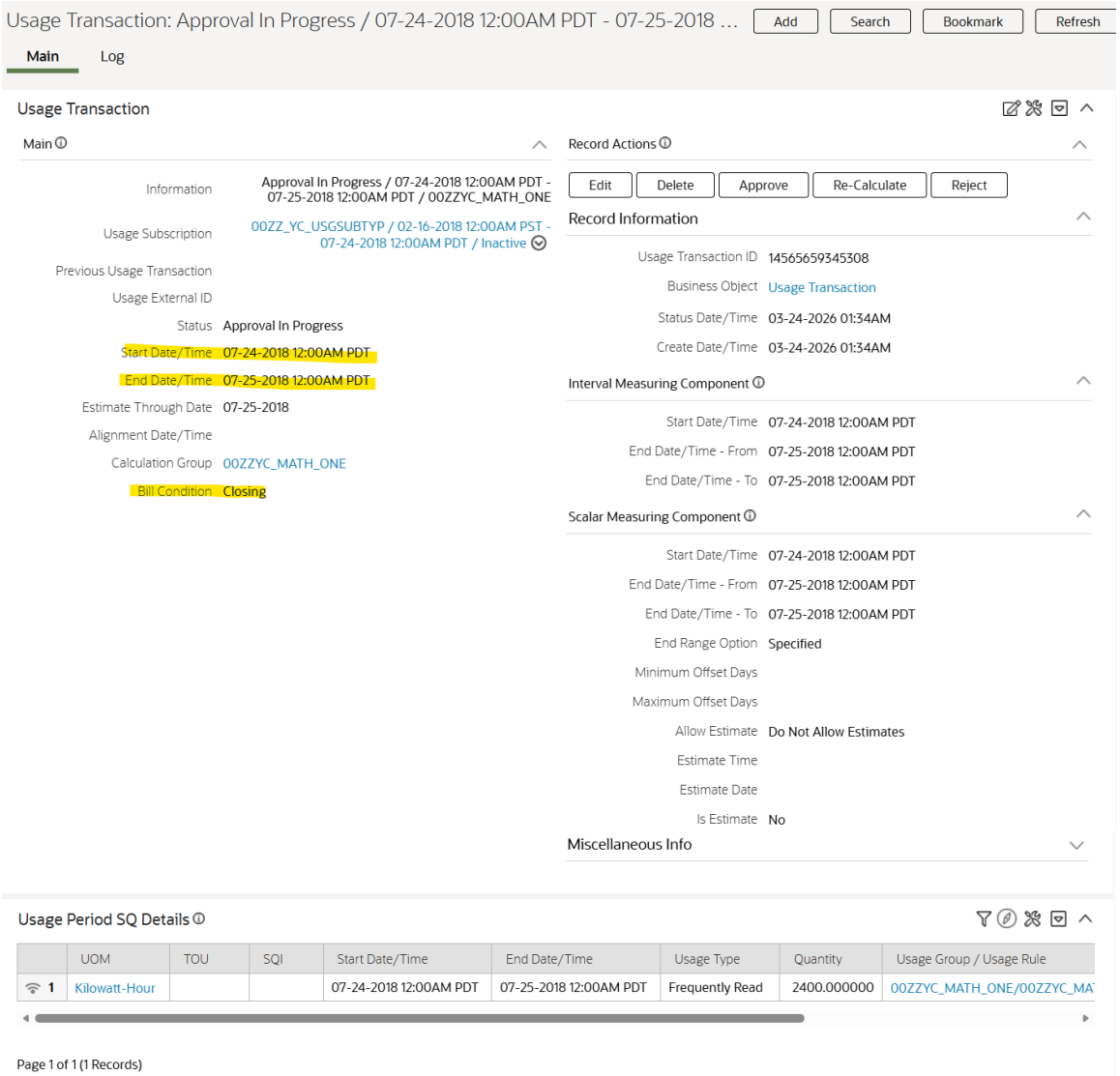
Usage Transaction ID: 14565659345308



Test Case 2: Verify that usage is calculated for a closing usage transaction that is created for two days.

1. Create a device, device configuration, interval MC.
2. Create a service point and install the device on service point.
3. Create measurements on the interval MC from 03-30-2018 12 A.M to 07-25-2018 12:00AM.
4. Create usage calculation group with D2-Math Usage rule.

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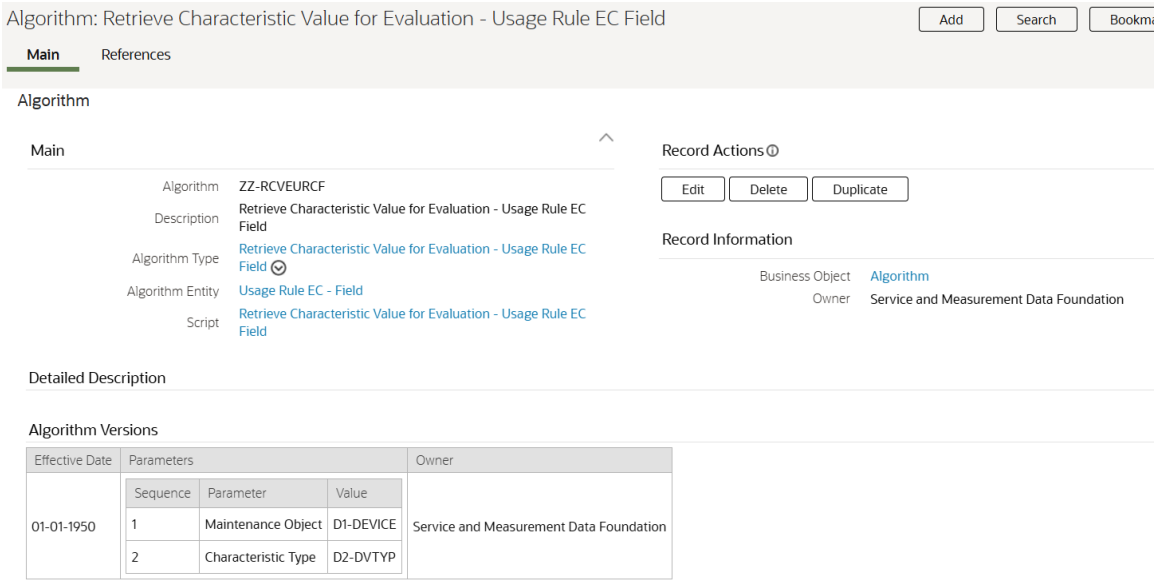
a. Add “Usage Calculation Rule Eligibility Criteria” using the Algorithm type D1-RCVEURCF (Retrieve Characteristic Value for Evaluation - Usage Rule EC Field).

Eligibility Criteria Algorithm



5. Create usage subscription with above service point and usage calculation group.
6. Move the usage subscription to inactive status and set the end date/time as 07-24-2018 12 A.M.
7. Create a usage transaction for the following period -

   Start date/time 07-23-2018 12 A.M and End date/time 07-25-2018 12 A.M.
8. Verify that usage is calculated on the usage transaction.

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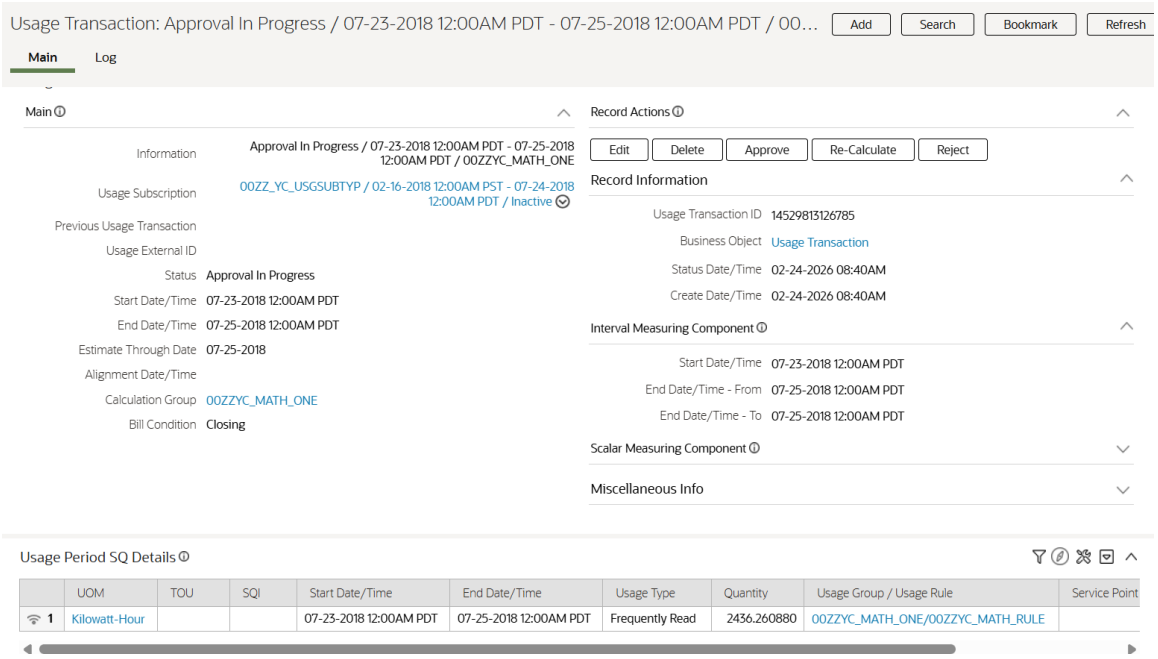

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